

HIMANSHU KIRAN GARUD

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EDUCATION

University of North Carolina at Charlotte

Master of Science in Computer Science

Aug 2023 - May 2025

GPA: 3.9/4.0

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, GoLang, C++, SQL, Linux.
Frontend: React, Vue.js, D3.js, ObservablePlot.js, HTML, CSS.
Backend Development: Node.js, Express, FastAPI, Flask, RESTful APIs, JWT Authentication, RBAC, Backend Validation, Stored Procedures.
Databases & Data: PostgreSQL, MySQL, MongoDB, Google BigQuery, Snowflake; Schema Design, Query Optimization, Semantic Views.
Cloud & Deployment: AWS (EC2, RDS, S3), GCP, Microsoft Azure (App Services, Static Web Apps), Docker, CI/CD Pipelines.
AI / Agentic Systems: Snowflake Cortex Agents, Cortex Search, Agentic AI, RAG Pipelines, LLM Tool Calling, GPT API Integration, Human-in-the-Loop Workflows.
ML / Systems: PyTorch, MediaPipe, Parallel Processing, Fault Tolerance, High Availability.

EXPERIENCE

Rebecca Everlene Trust Company

July 2025 - June 2026

Software Engineer

Remote, USA

- Built and deployed an **LLM-powered discovery platform** across **10K+ books** using **OpenAI APIs, PostgreSQL/pgvector, embeddings, semantic retrieval, metadata filtering, and LLM reranking** to deliver grounded recommendations through production backend services.
- Architected fault-tolerant services on **AWS EC2/RDS** with indexed queries, structured-output validation, retry mechanisms, and backend safeguards, reducing API latency by **35% to ~200ms** while maintaining reliability under concurrent traffic.

Electric Power Research Institute

June 2024 - May 2025

Software Engineer

Charlotte, NC

- Designed and deployed production-grade **Python FastAPI** data services on Azure with **CI/CD, RBAC, validation, and versioned ingestion** for **10K+** experimental records, supporting centralized engineering analytics workflows.
- Built interactive analytics workflows using **D3.js and ObservablePlot.js** for high-dimensional scientific datasets, reducing visualization update time by **20%** and enabling real-time exploration for research teams.
- Re-architected legacy engineering calculators from **Streamlit** into modular **Vue.js/TypeScript** components integrated into a centralized cloud platform, improving maintainability, scalability, and consistency across engineering applications.

Persistent Systems

May 2021 - July 2023

Software Engineer

India

- Architected a **Python/BigQuery data platform** for large-scale healthcare datasets, migrating **200+** legacy workflows into modular distributed services and reducing duplicated transformation logic by **25%**.
- Reduced end-to-end pipeline runtime from **8-9 hours to under 25 minutes** through **parallel processing, BigQuery query optimization, incremental execution, and automated failure recovery** across distributed production workloads.
- Refactored a legacy EdTech reporting backend by restructuring **100+** standalone SQL scripts into a shared ingestion-normalization-aggregation pipeline, eliminating **40%** duplicated logic and improving scalability of reusable backend components.
- Improved system stability and throughput by **20%** with automated validation, monitoring, and fault-tolerant mechanisms across distributed data workflows, ensuring consistent reporting and production data delivery.

Eastro Control Systems

Dec 2019 - Feb 2020

Software Engineer Intern

India

- Reduced response time by **30%** by engineering a **Flask-based** workforce scheduling system with **JWT authentication, RBAC, schema optimization, indexing, and query tuning** for efficient backend data retrieval.

University of North Carolina at Charlotte

Jan 2024 - May 2024

Research Assistant

Charlotte, NC

- Improved 3D human mesh reconstruction to **2.47 cm** localization accuracy by designing **CNN-based** models in **PyTorch** on mmWave radar point-cloud data, enabling real-time inference through a low-latency **UDP** streaming pipeline.

PROJECTS

RootCause AI - AI-Powered Developer Debugging Platform *FastAPI, Next.js, TypeScript, Ollama/Qwen, GitHub API, Docker Compose, React Flow*

- Built an evidence-backed **AI debugging platform** across **6 infrastructure technologies**, combining **FastAPI, Next.js, GitHub APIs, deterministic repository analysis, and LLM reasoning** for grounded root-cause diagnosis.
- Engineered a **repository-aware diagnostic pipeline** that parses **Docker Compose/configuration**, constructs dependency graphs, and returns **Pydantic-validated structured diagnoses** with confidence scores, remediation steps, and verification workflows.

SnowOps AI - Snowflake Operational Incident Agent *Snowflake Cortex Agents, Cortex Search, Semantic Views, SQL, Agentic AI, RAG*

- Built an **Agentic AI operational-agent POC** using **Snowflake Cortex Agents, Semantic Views, Cortex Search, RAG, and LLM tool calling** to perform automated root-cause analysis with approval guardrails, failure-state validation, and auditable retry actions.

ACHIEVEMENTS

- 1st Place**, Hack with CAIR Hackathon, UNC Charlotte - Built Posturelytics.AI, an AI-powered posture and focus tracker.
- Top 20 Finalist**, HooHacks 2025 (**800+** participants) - Recognized for innovation among 200+ ML-driven projects.
- High Five Award**, Persistent Systems - Honored for proactive ownership and impact in production environments.
- Microsoft Certified: **Azure AI Engineer Associate** (AI-102) - Passed with distinction.